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Mathematics Syllabus – S1-S3

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1. General Objectives

The European Schools have the two objectives of providing formal education and of encouraging pupils' personal development in a wider social and cultural context. Formal education involves the acquisition of competences (knowledge, skills and attitudes) across a range of domains. Personal development takes place in a variety of spiritual, moral, social and cultural contexts. It involves an awareness of appropriate behaviour, an understanding of the environment in which pupils live, and a development of their individual identity.

These two objectives are nurtured in the context of an enhanced awareness of the richness of European culture. Awareness and experience of a shared European life should lead pupils towards a greater respect for the traditions of each individual country and region in Europe, while developing and preserving their own national identities.

The pupils of the European Schools are future citizens of Europe and the world. As such, they need a range of competences if they are to meet the challenges of a rapidly-changing world. In 2006 the European Council and European Parliament adopted a European Framework for Key Competences for Lifelong Learning. It identifies eight key competences which all individuals need for personal fulfilment and development, for active citizenship, for social inclusion and for employment:

1. Literacy competence
2. Multilingual competence
3. Mathematical competence and competence in science, technology and engineering
4. Digital competence
5. Personal, social and learning to learn competence
6. Civic competence
7. Entrepreneurship competence
8. Cultural awareness and expression competence

The European Schools' syllabuses seek to develop all of these key competences in the pupils.

Key competences are that general, that we do not mention them all the time in the Science and Mathematics syllabuses.

2. Didactical Principles

General

In the description of the learning objectives, competences, connected to content, play an important role. This position in the learning objectives reflects the importance of competences acquisition in actual education. Exploratory activities by pupils support this acquisition of competences, such as in experimenting, designing, searching for explanations and discussing with peers and teachers. In science education, a teaching approach is recommended that helps pupils to get acquainted with concepts by having them observe, investigate and explain phenomena, followed by the step to have them make abstractions and models. In mathematics education, investigations, making abstractions and modelling are equally important. In these approaches, it is essential that a maximum of activity by pupils themselves is stimulated – not to be confused with an absent teacher: teacher guidance is an essential contribution to targeted stimulation of pupils' activities.

The concept of *inquiry-based learning* (IBL) refers to these approaches. An overview of useful literature on this can be found in the *PRIMAS guide for professional development providers*.

http://primas-project.eu/wp-content/uploads/sites/323/2017/10/PRIMAS_Guide-for-Professional-Development-Providers-IBL_110510.pdf

Mathematics

Careful thought has been given to the content and the structure to where topics are first met in a pupil's time learning mathematics in secondary education. It is believed that this is a journey and if too much content is met at one point, there is a risk that it will not be adequately understood and thus a general mathematical concept will not be fully appreciated. By limiting the content of this syllabus (found in section 4.2.) each year more time can be used to develop core mathematical concepts that may have been met before or new mathematical concepts introduced are given ample time for extension. It must be noted that extension activities are conducted at the discretion of the teacher, however, it is suggested that rather than look at a vertical approach to extension a horizontal approach is used, thus giving the pupil a deeper understanding of the mathematical concept (in section 4 the word 'limitation' is used to ensure the extension does not go too far).

Furthermore, to this point it is believed that with a focus on competences this syllabus can encourage pupils to have a greater enjoyment of mathematics, as they not only understand the content better but understand the historical context (where it is expected a history of mathematics can be told over the cycles) and how the mathematics can be applied in other subjects, cross cutting (these can be seen in the fourth column in section 4.2.). As such the syllabuses have specifically been designed with reflection to the key competences (section 1) and the subject specific competences (section 3.1). In some cases, the key competences are clear for example the numerous history suggested activities (shown by the icon) that maps to key competency 8 (Cultural awareness and expression). In other areas the link may not be so apparent.

One of the tasks in the pupil's learning process is developing inference skills, analytical skills and strategic thinking, which are linked to both the key and subject specific competences. This is the ability to plan further steps in order to succeed solving a problem as well as dividing the process of solving more complex problems into smaller steps. A goal of teaching mathematics is to develop pupil's intuitions in mathematics appropriate for their age. The ability to understand and use mathematical concepts (e.g. angle, length, area, formulae and equations) is much more important than memorising formal definitions.

This syllabus has also been written so that it can be accessible by teachers, parents and pupils. This is one reason why icons have been used (listed in section 4.2.). These icons represent different areas of mathematics and are not necessarily connected to just one competency but can cover a number of competences.

To ensure pupils have a good understanding of the mathematics the courses from S1 to S7 have been developed linearly with each year the work from the previous year is used as a foundation to build onto. Thus, it is essential before commencing a year the preceding course must have been covered or a course that is similar. The teacher is in the best position to understand the specific needs of the class and before beginning a particular topic it is expected that pupils have the pre-required knowledge. A refresh is always a good idea when meeting a concept for the first time in a while. It should be noted that revision is not included in the syllabus, however, as mentioned earlier about limiting new content, there is time to do this when needed.

The use of technology and digital tools plays an important role in both theoretical and applied mathematics, which is reflected in this syllabus. The pupils should get the opportunity to work and solve problems with different tools such as spreadsheets, computer algebra system (CAS) software, dynamic geometric software (DGS), programming software or other software that are available in the respective schools. Technology and digital tools should be used to support and promote pupils' understanding, for example by visualising difficult concepts and providing interactive and personalised learning opportunities, rather than as a substitute for understanding. Their use will also lead to improved digital competence.

Teachers have full discretion with how to teach this course, materials to use and even the sequence the content is taught in. The content and the competencies (indicated in the tables in section 4.2., columns 2 and 3) to be covered is, however, mandatory.

The S1 Course

The S1 course is a transitional year from primary mathematics to secondary in that tangible mathematical concepts first met in primary will become increasingly more abstract in future years in secondary. However, it is important that the challenge encountered from P5 to S1 is as smooth as possible. Thus, many tangible ideas are presented as activities in this first year. This is to further cement a full understanding of a mathematical concept.

It is expected with a focus on limiting the new mathematical concepts and building on core ideas met in primary that there is plenty of time for extension activities. An example of this could be when considering sequences. Rather than looking at nonlinear sequences after first meeting linear sequences have pupils try and discover linear sequences in everyday life.

With this in mind the idea of functions, historically an important topic in the European Baccalaureate, will be introduced in S1. This is first introduced using pictorial sequences, discovering a simple rule that generates the next term, developed using a table of values, using this to find a particular term in the sequence and then further developed by plotting these sequences. Using this tangible approach in S1 should lay the foundation for the more abstract ideas met in future years in sequences.

The S2 Course

In the specific case of the S2 course mathematics at this level is at a more formal level where many models can be used to give pupils (visual) support, e.g. the area model, a ratio table, tables as intermediary tool for creating a graph of a given situation.

In the S2 curriculum, still much attention is given for the Number topic and pupils' understanding

of the number system. More time is scheduled to develop their algebraic skills, where investigations of number and Geometric patterns play an important role, and pupils move from word formulae to formulae using a variable. As can be seen clearly here the mathematics is becoming more abstract compared to S1.

The investigations of 2D and 3D shapes are a base for the development of more formal mathematics concepts in the years that follow. When it comes to formula it is more important that a pupil understands which formula to apply and why the formula works rather than memorising the formula.

The S3 Course

In the specific case of the S3 pupils, mathematics reaches a more formal level, especially in the Algebra and Geometry topics. However, some pupils may be unable to express the correct solution in a formal way. It might be helpful to refer to models that can be used to give pupils (visual) support, e.g. the area model, a ratio table, tables as an intermediary tool for creating a graph of a given situation, Geometric software for making constructions.

Pupils who can easier work at a former level should be challenged to solve more complex problems. Their abilities may indicate at what level a pupil may be able to continue next year.

In the S3 curriculum, there is a little less attention for the Number strand, because many computations are integrated in the Algebra and Geometry strand.

The Statistics and Probability strand starts this year with counting problems: number of paths and arrangements. In Year 4 and 5, this will be continued and formalised in the topic probability.

3. Learning Objectives

3.1. Competences

The following are the list of subject specific competences for mathematics. Here the key vocabulary is listed so that when it comes to reading the tables in section 4.2. the competency being assessed can be quickly seen. Please note that the list of key vocabulary is not exhaustive, and the same word can apply to more than one competency depending on the context.

Further information about assessing the level of competences can be found in section 5.1. Attainment Descriptors. The key concepts here are those needed to attain a sufficient mark.

	Competency	Key concepts (attain E)	Key vocabulary
1.	**Knowledge and comprehension**	Demonstrates satisfactory knowledge and understanding of straightforward mathematical terms, symbols and principles	Apply, classify, compare, convert, define, determine, distinguish, expand, express, factorise, identify, know, manipulate, name, order, prove, recall, recognise, round, simplify, understand, verify
2.	**Methods**	Carries out mathematical processes in straightforward contexts, but with some errors	Apply, calculate, construct, convert, draw, manipulate model, organise, plot, show, simplify sketch solve, use, verify
3.	**Problem solving**	Translates routine problems into mathematical symbols and attempts to reason to a result	Analyse, classify, compare, create, develop, display, estimate, generate, interpret, investigate, measure, model, represent, round, simplify, solve
4.	**Interpretation**	Attempts to draw conclusions from information and shows limited understanding of the reasonableness of results	Calculate, conduct, create, develop, discover, display, generate, interpret, investigate, model
5.	**Communication**	Generally presents reasoning and results adequately; using some mathematical terminology and notation	Calculate, conduct, create, discover, display, interpret, investigate, model, present
6.	**Digital competence[1]**	Uses technology satisfactorily in straightforward situations	Calculate, construct, create, display, draw, model, plot, present, solve

1 This competence is part of the European Digital Competence Framework (https://ex.europa.eu/jrc/en/digcomp)

3.2. Cross-cutting concepts

Cross cutting concepts will be carried by the joint competences. The list of cross cutting concepts that will be composed will be shared by all science and mathematics syllabuses. The tentative list to be taught is based on the next generation science standards in the United states (National Research Council, 2013):

	Concept	Description
1.	**Patterns**	Observed patterns of forms and events guide organisation and classification, and they prompt questions about relationships and the factors that influence them.
2.	**Cause and effect**	Mechanism and explanation. Events have causes, sometimes simple, sometimes multifaceted. A major activity of science is investigating and explaining causal relationships and the mechanisms by which they are mediated. Such mechanisms can then be tested across given contexts and used to predict and explain events in new contexts.
3.	**Scale, proportion and quantity**	In considering phenomena, it is critical to recognise what is relevant at different measures of size, time, and energy and to recognise how changes in scale, proportion, or quantity affect a system's structure or performance.
4.	**Systems and system models**	Defining the system under study—specifying its boundaries and making explicit a model of that system—provides tools for understanding the world. Often, systems can be divided into subsystems and systems can be combined into larger systems depending on the question of interest
5.	**Flows, cycles and conservation**	Tracking fluxes of energy and matter into, out of, and within systems helps one understand the systems' possibilities and limitations.
6.	**Structure and function**	The way in which an object or living thing is shaped and its substructure determine many of its properties and functions and vice versa.
7.	**Stability and change**	For natural and built systems alike, conditions of stability and determinants of rates of change or evolution of a system are critical for its behaviour and therefore worth studying.
8.	**Nature of Science**	All science relies on a number of basic concepts, like the necessity of empirical proof and the process of peer review.
9.	**Value thinking**	Values thinking involves concepts of justice, equity, social–ecological integrity and ethics within the application of scientific knowledge.

In the mathematics syllabuses, the concepts 5 and 8 will be addressed only to a limited extent. The lists of competences and cross cutting concepts will serve as a main cross-curricular binding mechanism. The subtopics within the individual syllabuses will refer to these two aspects by linking to them in the learning goals.

http://ngss.nsta.org/Professional-Learning.aspx

4. Content

4.1. Topics

This section contains the tables with the learning objectives and the mandatory content for the strand Mathematics in S1, S2 and S3 (4 periods per week) cycle.

4.2. Tables

How to read the tables on the following pages

The learning objectives are the curriculum goals. They are described in the third column. These include the key vocabulary, highlighted in bold, that are linked to the specific mathematics competences found in section 3.1. of this document.

These goals are related to content and to competences. The mandatory content is described in the second column. The final column is used for suggested activities, key contexts and phenomena. The teacher is free to use these suggestions or use their own providing that the learning objective and competencies have been met.

Please note that the word ‘limitation’ is used to ensure that when extension is planned it is planned with the idea of horizontal extension rather than vertical extension as mentioned in section 2. of this document.

Use of icons

Furthermore, there are six different icons which indicate the areas met in the final column:

	Activity
	Cross-cutting concepts
	Digital competence
	Extension
	History
	Phenomenon

Each of these icons highlight a different area and are used to make the syllabus easier to read. These areas are based on the key competences mentioned in section 1 of this document.

Year S1

YEAR S1	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Numbers	Natural and integer numbers	**Understand** the difference between natural and integer numbers		How negative numbers came around due to money and the idea of debt
	Plotting points	**Plot** points on a plane (only with integer coordinates)		Introduce a CAD software
	Absolute value	**Understand** absolute value of an integer number and in particular negative number in relation to the number line using notation, e.g. $\lvert -3 \rvert = 3$		
	Comparison of integer numbers	**Know** how to compare two integer numbers		Heights on a map
	Order a set of integer numbers	**Order** a set of integer numbers		Loans/debts and savings
		Use the transitivity property of > and <		
	Prime numbers, factors, divisors	**Understand** prime numbers using the concept of factors and divisors		
		Understand that all natural numbers can be written as a product of primes (prime factor decomposition), e.g. $24 = 2 \times 2 \times 2 \times 3$		Use powers, e.g. $24 = 2^3 \times 3$
		Know how to write a natural number as a product of prime numbers		

YEAR S1	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
		Know the criteria of divisibility by 2, 3, 5 and 10		• Criteria of divisibility by 9, 11, 25, 50 and 100 • Sieve of Eratosthenes to determine prime numbers, e.g., up to 100, or cryptography
		Use prime numbers to find the Lowest Common Multiple (LCM) and the Highest Common Factor (HCF) for up to three numbers, e.g. up to find LCM and HCF of 24, 36 and 42		Worded problem, e.g. 3 runners are running around a 400 metres track. One is running at 100 metres every 30 seconds, one is running 100 metres every 45 seconds and one is running every 60 seconds. After what times do they meet?
Operations	Add and subtract	**Calculate** by adding and subtracting integers and decimals		Apply on a calculator
	Multiply and divide	**Calculate** by multiplying and dividing two integers		Apply on a calculator
	0 and 1	**Understand** the importance of 0 and 1		History of the 0 (India)
	Order of operations	**Apply** the order of operations rules, brackets included, to make calculations		The area model can be introduced and used here, also for calculating 12×23
Fractions	Introduction to fractions	**Understand** a fraction is a ratio of two integers		Importance of fractions – historically for sharing food
		Understand that different fractions can be equivalent and simplify fractions		Show that two ratios of different numbers can be equivalent, e.g. $\frac{8}{12} = \frac{10}{15}$
		Know how to convert a fraction to a decimal (terminating decimals) and		Find the functions on a calculator

YEAR S1	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
		vice-versa **Order** fractions and decimals into order of size and place them on the number line		

YEAR S1	TOPIC: ALGEBRA			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Formulae, tables, graphs	Relationships in patterns (sequences), tables and graphs	**Use** a pictorial sequence (arithmetic only)		Pupils to draw the next pattern, e.g. using matches:
		Create rules to determine the next term of an arithmetic sequence		The rule is ‘each time I add 4’
		Understand how to create a general rule for an arithmetic sequence		The general rule is $4n + 1$
		Know how to substitute into a general rule to find the value of any term		e.g. Find the 100th step
		Know how to create a table of values from a pictorial sequence or a list of numbers		Use a spreadsheet to help find rules and determine an nth term
		Plot the sequence from a table of values		Use an appropriate spreadsheet software
Expressions	Linear expressions	**Simplify** equivalent expressions, e.g. $a + a + a = 3a$		
Equations	Linear equations	**Solve** simple linear equations up to the form $ax + b = cx + d$ where a, b, c and d are integer numbers and x (unknown) is a rational number		Use CAS software or photo-recognise software on smartphones to solve equations

YEAR S1	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
3D	Top and side views of 3D shapes	**Recognise**, **draw** or **sketch** top and side views of 3D shapes (prisms, cylinders, pyramids, cones and spheres)		Investigate and compare the similarities and differences between 3D models • Use objects built of small cubes • Create different solids with a given volume
	Classify 3D shapes	**Classify 3D shapes** according various criteria: • faces, edges, vertices • parallel and perpendicular faces and edges • curved or flat faces		‘Euler’s formula’ of Platonic solids
	Draw 3D shapes	**Draw 3D shapes** using CAD software and by hand (only cubes and cuboids by hand)		
	Nets	**Recognise and construct** nets (prisms and pyramids)		Pupils bring in cereal boxes into class to investigate different types of packaging
2D	2D shapes	**Recognise** and **name** the following 2D shapes: • Triangles (equilateral, isosceles, scalene and right angled) • Quadrilaterals (square, rectangle, rhombus, parallelogram, kite and trapezium) • Regular polygons up to decagon • Circle		Find those structures in the real world Try to make a floor tiling out of the shapes

YEAR S1	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
	Classify 2D shapes	**Classify 2D shapes** with respect to: • parallelism • perpendicularity • or equality of sides		
	Lines and angles	**Know** how to measure and construct line segments and angles using protractor and set-square: • parallel lines, perpendicular lines • perpendicular bisectors of line segments • angles of given size		Search and apply the similar buttons in a CAD software
	Triangles	**Know** how to construct triangles given the following: • With all three sides (SSS) • When given one angle and two sides (SAS and SSA) • When given two angles and one side (ASA)	 	• Equipment to be used: ruler, pair of compasses and protractor • Using straws of different lengths, investigate whether or not three straws create a triangle • Investigate if more than one or even no triangle is possible when given two sides and an angle • Explore the sum of angles and investigate which properties the given angles have to have
Orientation and navigation	Compass directions	**Use** compass directions and distances to find a location		Use with maps
	Concept of an angle	**Develop** the concept of an angle by making turns over a given angle		e.g. a turn from N to E is 90 degrees

YEAR S1	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
		Understand half a turn, ¼ of a turn, …		Investigate the result of the sum of the angles of a triangle, even of triangles on the globe where sum is 270°
Measurements	Geometrical object and its measure	**Understand** the distinction between a geometrical object and its measure, e.g.: • a segment and its length • an angle and its size • a surface and its area • a solid and its volume		Measurement of the world, units
	Length	**Estimate** and **measure** lengths, using appropriate units for length when solving practical problems		Different unit systems
				Convert between different unit systems, e.g. metric and imperial units, ships and aeroplanes
	Concept of area	**Develop** the concept of area by estimation strategies using unit squares		Fits small squares in a big square
	Perimeter and area	**Recall** and **apply** appropriate formula to **calculate** perimeters and areas of squares and rectangles		Calculate perimeters and areas of compound shapes
	Concept of volume	**Develop** the concept of volume by using counting strategies to find out how many unit cubes fit in a cuboid		Fits small cubes in a big cube
	Conversion of units	**Know** and **convert** units (prefix and units) of lengths and mass (from milli		Very small and large units in physics or informatics, Nano, Giga, ….

YEAR S1	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
		up to and including kilo)		

YEAR S1	TOPIC: SET THEORY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Sets	Basic concepts	**Understand** basic concepts of a set, elements of a set, universal set, empty set, union, intersection and complement of a set		Set of evens and odds, set of primes
	Venn diagrams	**Draw** Venn diagrams, place elements in appropriate areas and use Venn diagrams to show logical relationships		Venn diagrams could be used to show the relationships between quadrilaterals with different properties
	Symbols	**Use** appropriately the symbols $\in$, $\notin$, $\cap$, $\cup$ and $\subset$		

Year S2

YEAR S2	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Rounding numbers	Rounding numbers	**Round** the answer to a certain number of decimals or a number of significant figures		Make the difference between accuracy of measurement and precision in physics
Fractions	Equivalent fractions	**Convert** two fractions to equivalent fractions with the same denominator		Slicing a cake or a pizza
Operations	Add and subtract	**Calculate** by adding and subtracting rational numbers		
	Multiply and divide	**Calculate** by multiplying and dividing rational numbers		Apply to real world solutions, e.g. worded questions
	Powers	**Understand** the concept of factors and **use** the power notation with a positive base		Investigate the similarities and differences between the shorter notations, e.g. for $4+4+4$ and $4\times4\times4$
		Recall and **use** the following power rules: • $a^m \times a^n = a^{m+n}$ • $a^m \div a^n = a^{m-n}$ • $(a^m)^n = a^{m\times n}$ • $(a\times b)^m = a^m \times b^m$ • $(a\div b)^m = a^m \div b^m$ *Limitation: m and n natural numbers*		Simplify expressions with powers
	Order of operations	**Apply** the order of operations rules, brackets and powers included, to make calculations		Apply the rules on calculators and try them on simple calculator apps on a smartphone

YEAR S2	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Percentages	Fractions and percentages	**Understand** and **know** the relationship between fractions and percentages: $1/100$ part of a number is the same as 1% of that number		History of percentage notation and the % sign
		Discover relationships between fractions and percentages		
		Understand and **use** the relationship between percentages and fractions to find a percentage of a number		Apply to real world situations, e.g. worded questions
Ratios	Equivalent ratios	**Generate** equivalent ratios supported by a context, e.g. recipes, **using** a ratio table		Ratios in daily life; '… per …' problems
	Scale line	**Construct** a scale line for a map or object		
	Scale drawing	**Construct** scale drawings		Scale line; ratio table
	Scale ratio	**Understand** the scale ratio of a model and maps		Investigate the scale ratio of toy models, and of pictures in a Biology book
				Project with Integrated Science: create a model of the sun-earth-moon-system
Ratios and proportions	Ratios and proportions	**Solve problems** where ratios (part to part) or proportions (part to whole) are involved		In context situations, compare ratios

YEAR S2	TOPIC: ALGEBRA			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Formulae, tables, graphs	Graphs	**Interpret** a time-distance graph		Review and extend from the discrete model seen in S1 to a continuous model now
	Table, graph	**Create** a graph from a table and vice versa		Investigate linear relations from economics, e.g. total cost and the price per item
	Linear situations	**Investigate** and **interpret** linear situations by a given graph or table		Use direct proportion $y = kx$, e.g. converting units, currency
	Quadratic situations	**Investigate** and **interpret** quadratic situations by a given graph or table		e.g. circumference and area of ‘growing’ squares
	Graphs	**Recognise** and **plot** graphs of linear and quadratic relationships of one variable		Plot with a CAD system
	(word) formula, graphs, tables	**Model** situations or procedures by translating them into (word) formulae and by using tables and/or graphs		Use formulae, graphs and tables as tools to model, investigate and compare situations, e.g. the total cost when you pay a base amount and an amount per time unit (salary, rent, phone call)
Expressions, equations	Expressions	**Understand** when and how algebraic expressions can be simplified, by combining like terms, by expanding a single bracket expression such as $a(b + c)$		The area model can be used to expand brackets
	Linear equations	**Solve** simple linear equations up to the form $ax + b = cx + d$ where a, b, c, d and x (unknown) are rational numbers		Extending to simplifying expressions such as $a(bx + c) + d(ex + f)$ where a, b, c, d, e, f and x are rational numbers

YEAR S2	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
2D	Quadrilaterals and triangles	**Investigate**, **recognise**, **classify** and **name** quadrilaterals and triangles with respect to: • Parallel sides • Perpendicular sides • Equality of sides		
	Lines and angles	**Construct** lines and angles using ruler and compass: • Parallel lines, perpendicular lines • Perpendicular bisectors of lines • Bisect an angle		• Using paper folding to create these lines and shapes • Finding the loci that is equidistant from two points, e.g. using a map Euler straight line Construct these lines and angles with an appropriate technological tool
	Symmetry	**Understand** the concept of symmetry **Identify** symmetry in shapes, the line(s) of reflection, the centre of symmetry		Rangoli and Escher artworks Symmetry in nature (faces, organs, mirrors, …)
Measurement	Quadrilaterals and triangles	**Recall** and **apply** appropriate formula to **calculate** the area of quadrilaterals (including those met in S1 and the following: rhombus, parallelogram, kite and trapezium) and triangles		Calculate perimeters and areas of compound shapes

YEAR S2	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
	Circle	**Recall** and **apply** appropriate formula to **calculate** the perimeter of a circle and area of a disk using π		Investigation of the ratio between diameter and circumference History of π
	Area and volume	**Understand** how to convert units for area (mm^2, cm^2, dm^2, m^2) **Understand** how to convert units for volume (mm^3, cm^3, dm^3, m^3; ml, cl, dl, l, hl and 1 dm^3 = 1 litre) **Use** appropriate units for area and volume when solving practical problems		Investigations of surface areas of packages/boxes Investigations of volumes of packages/boxes Investigate the relationship of area and volume of enlarged or reduced shapes Use the outcome to explain why small children quicker cool off (or warm up) than adults or mice than elephants

YEAR S2	TOPIC: STATISTICS AND PROBABILITY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Descriptive statistics	Collect data	**Set up** and **carry out** a simple survey to collect data **Use** tallying and frequency tables grouping data together	 	• Experiments with temperature over time, display the data, draw conclusions • Investigations: collect data, display the data, draw conclusions • Survey • Appreciate the difference between discrete and continuous data Collect data in Science experiments
	Diagrams	**Display** (collected) data**, construct** and **interpret** statistical diagrams • line graphs • frequency diagrams • pie charts		
	Measures of tendency	**Investigate** two sets of data making up measures to **compare** and make **conclusions**		Pupils may informally use the idea of mean, median, mode, range, …)

Year S3

YEAR S3	TOPIC: NUMBERS			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Operations	Roots	**Understand** and **use** the square root of a number		Make connections between the area of a square and its side length
		Use a calculator to find approximations of square roots		Cube roots (make connections between the volume of a cube and its side length)
	Powers	**Understand** and **use** powers with a negative base		Pico, Nano, Micro, Mega, Giga, Tera, Peta
	Scientific notation	**Understand** and **use** the scientific notation		Investigate the scientific notation on a calculator
	Order of operations	**Apply** the order of operations rules, brackets, powers and square roots included, to make calculations		Investigations how well a calculator can apply these rules
	Inverse	**Understand** the additive inverse and the multiplicative inverse of a number		
	Percentages and ratios	**Calculate** a percentage of a number, a percentage increase and decrease.		A percent bar can be used to give visual support
		Use the relationship between percentages and ratios to solve problems		Project with Integrated Science: design a healthy, varied and balanced menu for a week

YEAR S3	TOPIC: ALGEBRA			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Functions	Linear functions $y = ax + b$	**Identify** linear functions: Start to use the notation $x \rightarrow ax + b$		
	Graphs of linear functions	In situations **interpret** gradients and intercepts of graphs of linear functions graphically		Plot functions on a CAD software
	Direct and inverse proportion	**Solve problems** involving direct and inverse proportion, including graphical and algebraic representations		Use compound units such as speed, unit pricing and density to solve problems
Expressions, equations	Expressions	**Understand** the difference between an expression, an equation and a formula		
		Manipulate formula of the following type: • $a = bc$, $b = a/c$, • $a = b + c$, $b = a - c$, • $a = b^2$, $\lvert b \rvert = \sqrt{a}$	 	Circumference, area and volume Speed, density, …
		Understand when and how algebraic expressions can be manipulated to maintain equivalence by combining like terms, by expanding and simplify a bracket expression such as $a(b + c)$ and $(a + b)(c + d)$ and factorise $ab + ac$	 	The area model can be introduced to expand brackets a and b real numbers
		Expand and **factorise** • $(a + b)^2 = a^2 + 2ab + b^2$ • $(a - b)^2 = a^2 - 2ab + b^2$		Use these to calculate examples such as: • $103^2 = (100 + 3)^2$ • $99^2 = (100 - 1)^2$

YEAR S3	TOPIC: ALGEBRA			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
		• $(a+b)(a-b)=a^2-b^2$ where a and b are fractions and/or variables		• $51\times 49=(50+1)(50-1)$ a and b are real numbers
		Simplify expressions containing brackets, fractions and combinations of both		Simplify expressions containing even powers
	Equations	**Solve** linear equations up to the form $a(bx\pm c)=d(ex\pm f)$ where a,b,c,d,e,f and x (unknown) are rational numbers.		Use of a CAS software or formula recognition software

YEAR S3	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
2D	Angles	**Identify** equal angles: • vertically opposite angles • alternate interior and exterior angles • corresponding angles		Prove the result of the sum of the angles of a triangle
	Transformations	**Understand** and **apply** transformations • translations • reflections • rotations • enlargements (positive scale factor)		Introduce the concepts of scalar and vector quantities and how the properties are affected or not by the transformations Investigate how area is affected by an enlargement
Ratios in right angle triangles	Right angled triangles, ratios and angles	**Investigate** and **compare** the steepness of different slopes *Limitation: do not introduce trigonometric functions and avoid using sin, cos and tan as this will be met in future years.*		Who has designed the best paper plane? How can you compare them? Investigate how to determine the glide ratio of a paper plane Investigate and compare different glide ratios Construct right angled triangles to create a table that shows in the left column (glide) angles (10º, 20º, 30º, …) and in the right column the (glide) ratio of the two rectangle sides, written as a rate
		Use ratios to find the height of an object		The height of a tower is 50 meters. How long is its shadow on June 21st? And on December 21st? Find the height of an object given the angle of the sunrays and its shadow length (Eratosthenes)

YEAR S3	TOPIC: GEOMETRY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
				Find the circumference of the Earth (Eratosthenes)
Measurement	Area	**Recall** and **apply** appropriate formula to calculate surface areas of solids restricted to prisms		Calculate surface areas of cylinders
	Volume	**Recall** and **apply** appropriate formula to calculate volumes of solids restricted to prisms, cylinders, pyramids, cones and spheres		
	Area and volume	**Recall** and **apply** appropriate formula for area (see S2 syllabus) and volume to calculate length when given the area or volume		• Manipulate the formulae: recall and apply appropriate formula to calculate the area (face or base) when given a volume • Relations between area and volume, the most volume with the least surface area, …

YEAR S3	TOPIC: STATISTICS AND PROBABILITY			
Subtopic	**Content**	**Learning objective**	**Key contexts, phenomena and activities**	
Descriptive statistics	Mean, median, mode	**Calculate** the mean, the median and the mode from a data set **Develop** strategies and **use** the mean, the median and/or the mode to compare two sets of data to draw conclusions **Calculate** the mean, median and mode from a frequency table		Use data from scientific sources
Combinatorics	Number of paths, arrangements	**Use** systematic strategies for counting such as sample space diagrams and tree diagrams to list all outcomes *Limitation: do not use these for probability*		Count different arrangements, e.g. to choose a menu (2 starters, 3 main courses, 2 desserts), different routes to go from A to B on a grid or planning a seating arrangement *Limitation: avoid nCr/nPr and factorials*

5. Assessment

For each level there are attainment descriptors written by the competences, which give an idea of the level that pupils have to reach. They also give an idea of the kind of assessments that can be done.

With the competences are verbs that give an idea of what kind of assessment can be used to assess that goal. In the table with learning objectives these verbs are used and put bold, so there is a direct link between the competences and the learning objectives.

Assessing content knowledge can be done by written questions where the pupil has to respond on. Partly that can be done by multiple choice but competences as constructing explanations and engaging in argument as well as the key competences as communication and mathematical competence need open questions or other ways of assessing.

An assignment where pupils have to use their factual knowledge to make an article or poster about a (broader) subject can be used to also judge the ability to critically analyse data and use concepts in unfamiliar situations and communicate logically and concisely about the subject.

In Europe (and America) pupils must have some competence in designing and/or engineering (STEM education). So there has to be an assessment that shows the ability in designing and communicating. A design assessment can also show the ability in teamwork.

Pupils have to be able to do an (experimental) inquiry. An (open) inquiry should be part of the assessments. Assessing designing and inquiry can be combined with other subjects or done by one subject, so pupils are not obliged to do too many designing or open inquiry just for assessment at the end of a year.

Digital competence can be assessed by working with spreadsheets, gathering information from internet, measuring data with measuring programs and hardware, modelling theory on the computer and comparing the outcomes of a model with measured data. Do combine this with other assessments where this competence is needed.

Assessment is formative when either formal or informal procedures are used to gather evidence of learning during the learning process and are used to adapt teaching to meet student needs. The process permits teachers and students to collect information about student progress and to suggest adjustments to the teacher's approach to instruction and the student's approach to learning.

Assessment is summative when it is used to evaluate student learning at the end of the instructional process or of a period of learning. The purpose is to summarise the students' achievements and to determine whether, and to what degree, the students have demonstrated understanding of that learning.

For all assessment, the marking scale of the European schools shall be used, as described in "*Marking system of the European schools: Guidelines for use*" (Ref.: 2017-05-D-29-en-7).

5.1. Attainment Descriptors

	A	B	C	D	E	F	FX
	(9,0 - 10 Excellent)	(8,0 - 8,9 Very good)	(7,0 - 7,9 Good)	(6,0 - 6,9 Satisfactory)	(5,0 - 5,9 Sufficient)	(3,0 - 4,9 Weak / Failed)	(0 - 2,9 Very weak / Failed)
Knowledge and comprehension	Demonstrates comprehensive knowledge and understanding of mathematical terms, symbols and principles in all areas of the programme	Shows broad knowledge and understanding of mathematical terms, symbols and principles in all areas of the programme	Shows satisfactory knowledge and understanding of mathematical terms, symbols and principles in all areas of the programme	Shows satisfactory knowledge and understanding of mathematical terms, symbols and principles in most areas of the programme	Demonstrates satisfactory knowledge and understanding of straightforward mathematical terms, symbols and principles	Shows partial knowledge and limited understanding of mathematical terms, symbols, and principles	Shows very little knowledge and understanding of mathematical terms, symbols and principles
Methods	Successfully carries out mathematical processes in all areas of the syllabus	Successfully carries out mathematical processes in most areas of the syllabus	Successfully carries out mathematical processes in a variety of contexts	Successfully carries out mathematical processes in straightforward contexts	Carries out mathematical processes in straightforward contexts, but with some errors	Carries out mathematical processes in straightforward contexts, but makes frequent errors	Does not carry out appropriate processes
Problem solving	Translates complex non-routine problems into mathematical symbols and reasons to a correct result; makes and uses connections between different parts of the programme	Translates non-routine problems into mathematical symbols and reasons to a correct result; makes some connections between different parts of the programme	Translates routine problems into mathematical symbols and reasons to a correct result	Translates routine problems into mathematical symbols and reasons to a result	Translates routine problems into mathematical symbols and attempts to reason to a result	Translates routine problems into mathematical symbols and attempts to reason to a result only with help	Does not translate routine problems into mathematical symbols nor attempts to reason to a result

Interpretation	Draws full and relevant conclusions from information; evaluates reasonableness of results and recognises own errors	Draws relevant conclusions from information, evaluates reasonableness of results and recognises own errors	Draws relevant conclusions from information and attempts to evaluate reasonableness of results	Attempts to draw conclusions from information given, shows some understanding of the reasonableness of results	Attempts to draw conclusions from information and shows limited understanding of the reasonableness of results	Makes little attempt to interpret information	Does not interpret information
Communication	Consistently presents reasoning and results in a clear, effective and concise manner, using mathematical terminology and notation correctly	Consistently presents reasoning and results clearly using mathematical terminology and notation correctly	Generally presents reasoning and results clearly using mathematical terminology and notation correctly	Generally presents reasoning and results adequately using mathematical terminology and notation	Generally presents reasoning and results adequately; using some mathematical terminology and notation	Attempts to present reasoning and results using mathematical terms	Displays insufficient reasoning and use of mathematical terms
Digital competence	Uses technology appropriately and creatively in a wide range of situations	Uses technology appropriately in a wide range of situations	Uses technology appropriately most of the time	Uses technology satisfactorily most of the time	Uses technology satisfactorily in straightforward situations	Uses technology to a limited extent	Does not use technology satisfactorily

Annex 1: Suggested time frame

For this cycle, the following topics are described with only an estimated number of weeks to be reviewed by the teacher depending on the class.

Note: The designated weeks include assessments, time needed for practice and rehearsal, mathematics projects, school projects, etcetera.

Course	**S1**	**S2**	**S3**
Topic	**Weeks**		
Number	10	11	9
Algebra	7	9	10
Geometry	12	9	10
Statistics and Probability		3	3
Set theory	3		
Total	**32**	**32**	**32**